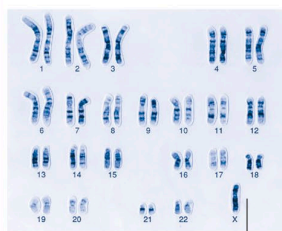


INHERITED DISORDERS

INHERITED DISORDERS ARE CAUSED BY DEFECTIVE GENES OR ABNORMAL CHROMOSOMES. IN CHROMOSOME DISORDERS, THERE IS A PROBLEM IN THE NUMBER OR STRUCTURE OF CHROMOSOMES, WHEREAS IN GENE DISORDERS, THERE IS A FAULT IN ONE OR MORE GENES.

CHROMOSOME DISORDERS

Two-thirds of chromosome disorders are numerical – egg or sperm cells have either too many or too few chromosomes. In many cases, they result in a miscarriage. In a few exceptions, the fetus survives. The most common is Down's syndrome, in which there is an extra chromosome 21. Abnormalities in the sex chromosomes have a less severe effect on the embryo, and there may not be any obvious signs of a problem. A girl with an extra X chromosome or a boy with an extra Y chromosome will probably go unnoticed. However, a boy who is born with an extra X chromosome (XXY) will have Klinefelter's syndrome, which becomes apparent at puberty when secondary sexual characteristics fail to develop. A girl who is born with only one X chromosome will have Turner's syndrome.



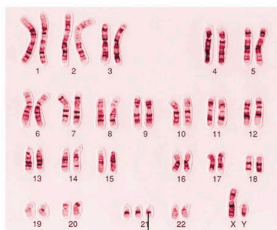
Missing X chromosome

TURNER'S SYNDROME

This chromosome set from a female shows only one X chromosome. She has Turner's syndrome, and will be short in stature and probably infertile.

DOWN'S SYNDROME

This chromosome set from a male shows an extra chromosome 21. He has Down's syndrome, and will have a distinct physical appearance and learning difficulties.



Extra chromosome 21

CYSTIC FIBROSIS

Cystic fibrosis is a gene disorder in which mucus glands produce abnormally thick secretions that cause repeated lung infections and problems digesting food. Weight gain is reduced, growth is slow, and life expectancy is shortened. Cystic fibrosis is caused by an abnormal gene that has to be received from each parent.

Prenatal genetic testing and genetic counselling will be offered to parents of one affected child if they want to have more.

Lungs

Lung infections, constant cough, and breathlessness

Pancreas

Lack of enzymes means digestion is inefficient

Intestines

Poor absorption of nutrients and intestinal blockage

Testes

Infertility as vas deferens and epididymis fail to develop properly

EFFECTS OF CYSTIC FIBROSIS

Many parts of the body are affected by excess mucus, causing poor health and slow physical development with intermittent episodes of serious illness.

